

Eigenvalues and Eigenvectors



Outline

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- Triangular and diagonal matrices

- Trace, determinant and eigenvalues

- Eigenvectors of distinct eigenvalues

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Eigenvalues

Introduction: Eigenvalues

- ▶ In this section we define the **eigenvalues** and the **eigenvectors** of a square matrix
- ▶ We will see how to compute them through the **characteristic equation** $\det(\mathbf{Q} - \lambda \mathbf{I}) = 0$
- ▶ For symmetric matrices, the signs of the eigenvalues tell us whether the matrix is (semi)definite: this is fundamental in optimization, where definiteness decides the nature of stationary points

Eigenvalue and eigenvector

Definition 1: eigenvalue and eigenvector

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be a square matrix.

*A scalar $\lambda \in \mathbb{R}$ is called an **eigenvalue** of $\mathbf{Q}$ if there exists a nonzero vector $\mathbf{v} \in \mathbb{R}^n \setminus \{0\}$ such that*

$$\mathbf{Q} \mathbf{v} = \lambda \mathbf{v}.$$

*In this case, $\mathbf{v}$ is called an **eigenvector** associated with λ .*

Characteristic equation

- ▶ Eigenvalues are solutions of the **characteristic equation**

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = 0.$$

- ▶ If $\mathbf{Q}$ is symmetric, then all its eigenvalues are real.
- ▶ The function $p(\lambda) = \det(\mathbf{Q} - \lambda \mathbf{I})$ is a polynomial of degree n in λ , called the **characteristic polynomial** of $\mathbf{Q}$.

Computing the eigenvectors

- ▶ If λ is an eigenvalue, the eigenvectors associated with λ are the nonzero solutions $\mathbf{v}$ of the homogeneous system

$$(\mathbf{Q} - \lambda \mathbf{I}) \mathbf{v} = \mathbf{0}.$$

- ▶ If $\mathbf{v}$ is an eigenvector associated with λ , then also $\alpha \mathbf{v}$ is an eigenvector associated with λ , for every $\alpha \in \mathbb{R} \setminus \{0\}$.
- ▶ Eigenvectors are therefore never unique: we just give one of them (“for example”).

Eigenvalues and definiteness

A square **symmetric** matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is

- ▶ **positive semidefinite** if and only if all its eigenvalues are non-negative.
- ▶ **positive definite** if and only if all its eigenvalues are strictly positive.
- ▶ **negative semidefinite** if and only if all its eigenvalues are non-positive.
- ▶ **negative definite** if and only if all its eigenvalues are strictly negative.

Eigenvalues and definiteness: remarks (1/2)

- ▶ Recall that (semi)definiteness has been defined in the chapter on matrices through the sign of $\mathbf{x}'\mathbf{Q}\mathbf{x}$: for instance, $\mathbf{Q}$ is positive semidefinite if $\mathbf{x}'\mathbf{Q}\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
- ▶ The previous slide gives an equivalent characterization in terms of eigenvalues.
- ▶ A symmetric matrix which is neither positive semidefinite nor negative semidefinite is called **indefinite**: equivalently, it has at least one strictly positive and at least one strictly negative eigenvalue.

Eigenvalues and definiteness: remarks (2/2)

- One direction is easy to see: if $\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq 0$, then

$$\mathbf{v}'\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}'\mathbf{v} = \lambda \sum_{i=1}^n v_i^2, \quad \text{with} \quad \sum_{i=1}^n v_i^2 > 0,$$

so $\mathbf{v}'\mathbf{Q}\mathbf{v}$ and λ have the same sign.

- For example, if $\mathbf{Q}$ is positive semidefinite, then every eigenvalue satisfies $\lambda \geq 0$.

Example (1/5)

Let us consider the matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The characteristic equation is:

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & -1 \\ -1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Example (2/5)

Hence

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = 0 \iff (\lambda - 1)(\lambda - 3) = 0,$$

and therefore the eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 3.$$

Example (3/5)

An eigenvector associated with $\lambda_1 = 1$ can be found by solving

$$(\mathbf{Q} - \mathbf{I})\mathbf{v} = 0 \iff \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{v} = 0,$$

which gives (for example)

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Example (4/5)

Similarly, an eigenvector associated with $\lambda_2 = 3$ satisfies

$$(\mathbf{Q} - 3\mathbf{I})\mathbf{v} = 0 \iff \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \mathbf{v} = 0,$$

which gives (for example)

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Therefore,

$$\mathbf{Q} \mathbf{v}_1 = 1 \mathbf{v}_1, \quad \mathbf{Q} \mathbf{v}_2 = 3 \mathbf{v}_2.$$

Example (5/5)

We can verify it directly:

$$\mathbf{Q} \mathbf{v}_1 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \mathbf{v}_1,$$

$$\mathbf{Q} \mathbf{v}_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} = 3 \mathbf{v}_2.$$

Since $\mathbf{Q}$ is symmetric and both eigenvalues are strictly positive, the matrix $\mathbf{Q}$ is **positive definite**.

Summary

Eigenvalues and eigenvectors:

- ▶ λ is an eigenvalue of $\mathbf{Q}$ if $\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \neq 0$
- ▶ Eigenvalues are the roots of the characteristic polynomial $p(\lambda) = \det(\mathbf{Q} - \lambda\mathbf{I})$
- ▶ Eigenvectors are the nonzero solutions of $(\mathbf{Q} - \lambda\mathbf{I})\mathbf{v} = 0$, and they are never unique

Definiteness of a symmetric matrix:

- ▶ Positive (semi)definite $\iff$ all eigenvalues > 0 (≥ 0)
- ▶ Negative (semi)definite $\iff$ all eigenvalues < 0 (≤ 0)
- ▶ Indefinite $\iff$ at least one positive and one negative eigenvalue

Properties of eigenvalues

Introduction: Properties of eigenvalues

- ▶ In this section we study some useful properties of eigenvalues
- ▶ For triangular and diagonal matrices the eigenvalues can be read directly on the diagonal
- ▶ The trace and the determinant of a matrix are linked to its eigenvalues: this gives quick checks of our computations
- ▶ Eigenvectors associated with distinct eigenvalues are linearly independent

Triangular and diagonal matrices

Observation 1: eigenvalues of triangular and diagonal matrices

Let $Q \in \mathbb{R}^{n \times n}$ be an upper triangular, a lower triangular or a diagonal matrix. Then the eigenvalues of Q are its diagonal entries $q_{11}, q_{22}, \dots, q_{nn}$.

Proof: Eigenvalues of triangular and diagonal matrices

- ▶ If $\mathbf{Q}$ is triangular (or diagonal), then also $\mathbf{Q} - \lambda \mathbf{I}$ is triangular (or diagonal), with diagonal entries $q_{11} - \lambda, \dots, q_{nn} - \lambda$.
- ▶ Recall that the determinant of a triangular matrix is the product of its diagonal entries. Hence

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \prod_{i=1}^n (q_{ii} - \lambda),$$

which is equal to zero if and only if $\lambda = q_{ii}$ for some $i \in \{1, 2, \dots, n\}$.



Example

Let us consider the upper triangular matrix

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 5 \end{pmatrix}.$$

The characteristic equation is

$$\begin{aligned} \det(\mathbf{U} - \lambda \mathbf{I}) &= \det \begin{pmatrix} 2 - \lambda & 1 & 3 \\ 0 & -1 - \lambda & 4 \\ 0 & 0 & 5 - \lambda \end{pmatrix} \\ &= (2 - \lambda)(-1 - \lambda)(5 - \lambda) = 0, \end{aligned}$$

and therefore the eigenvalues are the diagonal entries:

$$\lambda_1 = 2, \quad \lambda_2 = -1, \quad \lambda_3 = 5.$$

Trace of a matrix

Definition 2: trace of a matrix

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be a square matrix. The **trace** of $\mathbf{Q}$ is the sum of its diagonal entries:

$$\text{tr}(\mathbf{Q}) = \sum_{i=1}^n q_{ii} = q_{11} + q_{22} + \cdots + q_{nn}.$$

Roots of the characteristic polynomial (1/2)

- ▶ The characteristic polynomial $p(\lambda) = \det(\mathbf{Q} - \lambda \mathbf{I})$ has degree n .
- ▶ Over the complex numbers, a polynomial of degree n has exactly n roots $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$, provided that each root is counted as many times as its **multiplicity** (the number of times the factor $(\lambda_i - \lambda)$ appears in the factorization).
- ▶ Hence

$$p(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda).$$

Roots of the characteristic polynomial (2/2)

- ▶ The real roots are the eigenvalues of Definition 1.
- ▶ The roots which are not real are called **complex eigenvalues** of Q (their eigenvectors have complex entries).
- ▶ If Q is symmetric, all the roots are real.

Observation 2: determinant, trace and eigenvalues

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and let $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$ be the roots of its characteristic polynomial, each repeated according to its multiplicity. Then

$$\det(\mathbf{Q}) = \prod_{i=1}^n \lambda_i, \quad \operatorname{tr}(\mathbf{Q}) = \sum_{i=1}^n \lambda_i.$$

Proof: Determinant, trace and eigenvalues (idea)

Determinant: setting $\lambda = 0$ in $p(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ gives

$$\det(\mathbf{Q}) = p(0) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Trace: we compare the coefficients of λ^{n-1} .

- ▶ In the product $(\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ this coefficient is $(-1)^{n-1} \sum_{i=1}^n \lambda_i$.
- ▶ In $\det(\mathbf{Q} - \lambda \mathbf{I})$ the power λ^{n-1} only comes from the product of the diagonal entries $(q_{11} - \lambda) \cdots (q_{nn} - \lambda)$, and its coefficient is $(-1)^{n-1} \sum_{i=1}^n q_{ii}$.



Example (1/3)

For the matrix of the first example we have $\lambda_1 = 1$ and $\lambda_2 = 3$, and indeed

$$\operatorname{tr} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 2 + 2 = 4 = 1 + 3,$$

$$\det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 = 1 \cdot 3.$$

Example (2/3)

Multiplicities must be taken into account. The diagonal matrix

$$\mathbf{D} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

has characteristic polynomial $(4 - \lambda)^2(-3 - \lambda)$: the eigenvalue 4 has multiplicity 2 and the eigenvalue -3 has multiplicity 1.

Hence $\lambda_1 = \lambda_2 = 4$, $\lambda_3 = -3$, and

$$\operatorname{tr}(\mathbf{D}) = 4 + 4 - 3 = 5, \quad \det(\mathbf{D}) = 4 \cdot 4 \cdot (-3) = -48.$$

Example (3/3)

Complex roots must be taken into account. For the (non symmetric) matrix

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{we have} \quad \det(\mathbf{R} - \lambda \mathbf{I}) = \lambda^2 + 1,$$

which has no real roots: $\mathbf{R}$ has no (real) eigenvalues.

Its complex eigenvalues are $\lambda_1 = i$ and $\lambda_2 = -i$, and indeed

$$\operatorname{tr}(\mathbf{R}) = 0 = i + (-i), \quad \det(\mathbf{R}) = 1 = i \cdot (-i).$$

Linear independence

- Recall that the vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

implies $\alpha_1 = \dots = \alpha_k = 0$.

Eigenvectors of distinct eigenvalues

Observation 3: eigenvectors of distinct eigenvalues

Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be **distinct** eigenvalues of $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be associated eigenvectors ($\mathbf{Q}\mathbf{v}_i = \lambda_i\mathbf{v}_i$). Then $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent.

Eigenvectors of distinct eigenvalues: consequences

- ▶ In particular, if $\mathbf{Q} \in \mathbb{R}^{n \times n}$ has n distinct real eigenvalues, then one eigenvector for each eigenvalue gives n linearly independent vectors of $\mathbb{R}^n$.
- ▶ In the first example, the eigenvectors $\mathbf{v}_1 = (1, 1)'$ and $\mathbf{v}_2 = (1, -1)'$ are linearly independent, since neither of them is a multiple of the other.

Summary

Triangular and diagonal matrices:

- ▶ The eigenvalues are the diagonal entries $q_{11}, \dots, q_{nn}$

Trace, determinant and eigenvalues:

- ▶ $\text{tr}(\mathbf{Q}) = \sum_i q_{ii}$
- ▶ $\det(\mathbf{Q}) = \prod_i \lambda_i$ and $\text{tr}(\mathbf{Q}) = \sum_i \lambda_i$, counting multiplicities and complex roots

Distinct eigenvalues:

- ▶ Their eigenvectors are linearly independent

A complete example with a 3×3 matrix

Introduction: A complete example with a 3×3 matrix

- ▶ In this section we put together everything we have seen so far on a symmetric 3×3 matrix
- ▶ We compute the characteristic polynomial with the Laplace expansion, the eigenvalues and one eigenvector for each eigenvalue
- ▶ Finally, we check the results with the trace and the determinant, and we decide the definiteness of the matrix

Example (1/6)

Let us consider the symmetric matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Characteristic polynomial. Expanding along the first row (Laplace expansion):

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & 1 & 0 \\ 1 & 3 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{pmatrix}$$

Example (2/6)

$$\begin{aligned}\det(\mathbf{Q} - \lambda \mathbf{I}) &= (2 - \lambda)[(3 - \lambda)(2 - \lambda) - 1] \\ &\quad - 1 \cdot [1 \cdot (2 - \lambda) - 1 \cdot 0] \\ &= (2 - \lambda)[(3 - \lambda)(2 - \lambda) - 2] \\ &= (2 - \lambda)(\lambda^2 - 5\lambda + 4) \\ &= (2 - \lambda)(\lambda - 1)(\lambda - 4).\end{aligned}$$

Therefore the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 4.$$

Example (3/6)

Eigenvector for $\lambda_1 = 1$. We solve

$$(\mathbf{Q} - \mathbf{I})\mathbf{v} = 0 \iff \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \mathbf{v} = 0 \iff \begin{cases} v_1 + v_2 = 0 \\ v_1 + 2v_2 + v_3 = 0 \\ v_2 + v_3 = 0 \end{cases}$$

From the first and the third equation, $v_1 = -v_2$ and $v_3 = -v_2$ (the second equation is then satisfied). Choosing $v_2 = -1$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

Example (4/6)

Eigenvector for $\lambda_2 = 2$. We solve

$$(\mathbf{Q} - 2\mathbf{I})\mathbf{v} = 0 \iff \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix} \mathbf{v} = 0 \iff \begin{cases} v_2 = 0 \\ v_1 + v_2 + v_3 = 0 \end{cases}$$

hence $v_2 = 0$, $v_3 = -v_1$ and, for example,

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Example (5/6)

Eigenvector for $\lambda_3 = 4$. We solve

$$(\mathbf{Q} - 4\mathbf{I})\mathbf{v} = 0 \iff \begin{pmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{pmatrix} \mathbf{v} = 0$$

$$\iff \begin{cases} -2v_1 + v_2 = 0 \\ v_1 - v_2 + v_3 = 0 \\ v_2 - 2v_3 = 0 \end{cases}$$

From the first and the third equation, $v_2 = 2v_1$ and $v_3 = v_2/2 = v_1$ (the second equation is then satisfied: $v_1 - 2v_1 + v_1 = 0$).

Choosing $v_1 = 1$:

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

Example (6/6)

Checks. By Observation 2,

$$\operatorname{tr}(\mathbf{Q}) = 2 + 3 + 2 = 7 = 1 + 2 + 4,$$

$$\det(\mathbf{Q}) = 2 \cdot (6 - 1) - 1 \cdot (2 - 0) = 8 = 1 \cdot 2 \cdot 4.$$

Since the three eigenvalues are distinct, by Observation 3 the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent.

Finally, $\mathbf{Q}$ is symmetric with strictly positive eigenvalues, hence it is **positive definite**.

Summary

How to compute eigenvalues and eigenvectors:

- ▶ Compute $p(\lambda) = \det(\mathbf{Q} - \lambda \mathbf{I})$ (e.g., with the Laplace expansion) and factorize it
- ▶ The roots of $p(\lambda)$ are the eigenvalues
- ▶ For each eigenvalue λ , solve $(\mathbf{Q} - \lambda \mathbf{I})\mathbf{v} = 0$ and choose one nonzero solution

How to check the results:

- ▶ Sum of the eigenvalues = trace, product of the eigenvalues = determinant

Sylvester's criterion

Introduction: Sylvester's criterion

- ▶ Computing all the eigenvalues of a matrix can be difficult
- ▶ For symmetric matrices, definiteness and semidefiniteness can also be checked by computing determinants of suitable square submatrices: the **principal minors**
- ▶ In this section we introduce principal minors and Sylvester's criterion, and we see why for semidefiniteness the leading principal minors are **not** enough

Definition 3: principal minors

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$.

- ▶ A **principal minor** of order k of $\mathbf{Q}$ is the determinant of the $k \times k$ submatrix obtained from $\mathbf{Q}$ by keeping the rows and the columns with the **same** indices $i_1 < i_2 < \dots < i_k$ (equivalently, by deleting the same $n - k$ rows and columns).
- ▶ The **leading principal minor** of order k is the principal minor obtained by keeping the first k rows and columns:

$$\Delta_k = \det \begin{pmatrix} q_{11} & \cdots & q_{1k} \\ \vdots & \ddots & \vdots \\ q_{k1} & \cdots & q_{kk} \end{pmatrix}, \quad k = 1, 2, \dots, n.$$

Principal minors: remarks

- ▶ A matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ has exactly n leading principal minors

$$\Delta_1 = q_{11}, \quad \Delta_2, \quad \dots, \quad \Delta_n = \det(\mathbf{Q}).$$

- ▶ For $n = 3$, the principal minors are:
 - ▶ of order 1: q_{11}, q_{22}, q_{33} ;
 - ▶ of order 2: the determinants of the submatrices with indices $\{1, 2\}, \{1, 3\}, \{2, 3\}$;
 - ▶ of order 3: $\det(\mathbf{Q})$.

Theorem 1: Sylvester's criterion

Let $Q \in \mathbb{R}^{n \times n}$ be a **symmetric** matrix with leading principal minors $\Delta_1, \Delta_2, \dots, \Delta_n$. Then

- ▶ Q is **positive definite** if and only if

$$\Delta_k > 0, \quad \forall k \in \{1, 2, \dots, n\};$$

- ▶ Q is **negative definite** if and only if the leading principal minors alternate in sign starting with a negative one:

$$\Delta_1 < 0, \quad \Delta_2 > 0, \quad \Delta_3 < 0, \quad \dots$$

i.e.,

$$(-1)^k \Delta_k > 0, \quad \forall k \in \{1, 2, \dots, n\}.$$

Observation 4: semidefinite matrices and principal minors

Let $Q \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then

- ▶ Q is **positive semidefinite** if and only if **all** its principal minors (not only the leading ones) are ≥ 0 ;
- ▶ Q is **negative semidefinite** if and only if every principal minor of order k has the sign of $(-1)^k$ or is zero, i.e., $(-1)^k$ (principal minor of order k) ≥ 0 , for all k .

For semidefiniteness it is **not** enough to check $\Delta_k \geq 0$ for the leading principal minors only.

Example: a positive definite matrix

For the matrix of the complete 3×3 example

$$\mathbf{Q} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

the leading principal minors are

$$\Delta_1 = 2 > 0, \quad \Delta_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 6 - 1 = 5 > 0,$$

$$\Delta_3 = \det(\mathbf{Q}) = 8 > 0.$$

By Sylvester's criterion, $\mathbf{Q}$ is positive definite, in agreement with its eigenvalues 1, 2, 4.

Example: an indefinite matrix (1/2)

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

We have $\Delta_1 = 1 > 0$ and $\Delta_2 = 1 - 4 = -3 < 0$. Hence:

- ▶ $\mathbf{Q}$ is not positive definite ($\Delta_2 < 0$) and not negative definite ($\Delta_1 > 0$);
- ▶ $\mathbf{Q}$ is not positive semidefinite (the principal minor Δ_2 of order 2 is negative) and not negative semidefinite ($(-1)^2 \Delta_2 < 0$).

Therefore $\mathbf{Q}$ is **indefinite**.

Example: an indefinite matrix (2/2)

Indeed, recalling the definition via $\mathbf{x}'\mathbf{Q}\mathbf{x}$:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} : \quad \mathbf{x}'\mathbf{Q}\mathbf{x} = 1 > 0,$$

$$\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} : \quad \mathbf{x}'\mathbf{Q}\mathbf{x} = 1 - 4 + 1 = -2 < 0.$$

The eigenvalues are the roots of

$$(1 - \lambda)^2 - 4 = (\lambda - 3)(\lambda + 1),$$

i.e., 3 and -1 : one positive and one negative.

Example: a singular positive semidefinite matrix (1/2)

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The leading principal minors are

$$\Delta_1 = 1, \quad \Delta_2 = 2 - 1 = 1,$$

$$\Delta_3 = \det(\mathbf{Q}) = 1 \cdot (2 - 1) - 1 \cdot (1 - 0) = 0.$$

Since $\Delta_3 = 0$, $\mathbf{Q}$ is **not** positive definite.

Example: a singular positive semidefinite matrix (2/2)

To decide whether it is positive semidefinite we check **all** principal minors:

- ▶ order 1: $q_{11} = 1$, $q_{22} = 2$, $q_{33} = 1$;
- ▶ order 2: $\det \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = 1$, $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$, $\det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = 1$;
- ▶ order 3: $\det(\mathbf{Q}) = 0$.

All principal minors are ≥ 0 , hence $\mathbf{Q}$ is **positive semidefinite**.

Indeed

$$\mathbf{x}'\mathbf{Q}\mathbf{x} = x_1^2 + 2x_1x_2 + 2x_2^2 + 2x_2x_3 + x_3^2 = (x_1 + x_2)^2 + (x_2 + x_3)^2 \geq 0,$$

and the eigenvalues are 0, 1, 3 ($\mathbf{v} = (1, -1, 1)'$ satisfies $\mathbf{Q}\mathbf{v} = 0 = 0\mathbf{v}$).

Example: leading principal minors are not enough

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

The leading principal minors are $\Delta_1 = 0 \geq 0$ and $\Delta_2 = 0 \geq 0$. However $\mathbf{Q}$ is **not** positive semidefinite: for

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{we have} \quad \mathbf{x}' \mathbf{Q} \mathbf{x} = -1 < 0.$$

The principal minor $q_{22} = -1$ (which is not a leading one) detects this. In fact, the eigenvalues are 0 and -1 , and $\mathbf{Q}$ is negative semidefinite.

Summary

Principal minors:

- ▶ Determinants of the submatrices with the same row and column indices
- ▶ Leading principal minors $\Delta_1, \dots, \Delta_n$: first k rows and columns

Sylvester's criterion (symmetric matrices):

- ▶ Positive definite $\iff \Delta_k > 0$ for all k
- ▶ Negative definite $\iff (-1)^k \Delta_k > 0$ for all k
- ▶ Semidefiniteness: check **all** principal minors, not only the leading ones

Summary (1/2)

Eigenvalues and eigenvectors:

- ▶ $\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq 0$; eigenvalues are the roots of $\det(\mathbf{Q} - \lambda\mathbf{I})$
- ▶ Eigenvectors: nonzero solutions of $(\mathbf{Q} - \lambda\mathbf{I})\mathbf{v} = 0$

Properties:

- ▶ Triangular and diagonal matrices: eigenvalues on the diagonal
- ▶ $\det(\mathbf{Q}) = \prod_i \lambda_i$, $\text{tr}(\mathbf{Q}) = \sum_i \lambda_i$
- ▶ Distinct eigenvalues $\Rightarrow$ linearly independent eigenvectors

Summary (2/2)

Definiteness of a symmetric matrix:

- ▶ Through the signs of the eigenvalues
- ▶ Through Sylvester's criterion (leading principal minors) for definiteness
- ▶ Through all the principal minors for semidefiniteness